Validating LLM Reproducibility Evaluators: Agreement Degrades on AI-Improved Bug Reports

Tong Gia Huy, Tran Thai Duong, Nguyen Hong Quan, Nguyen Thanh Loc, Lam Song Toan
FPT University, Vietnam

Abstract

I. Abstract

Reproducing a reported bug is often the slowest step in triage, and judging whether a bug report contains enough information to be reproduced is a manual, time-consuming task performed by developers. Whether a large language model (LLM) can perform this judgment reliably enough to assist that process is untested: prior work uses LLMs to generate or improve bug reports, but rarely validates them as reproducibility *assessors* against human judgment. We evaluate GPT-4o mini as an automated reproducibility evaluator on the ImproBR replication package, which provides two forms of the same 139 Mojira Minecraft bug reports: the original (*Raw*) reports, and *Improved* versions in which an LLM pipeline has rewritten each report to be more complete. Each form was independently annotated by two human reviewers along three structured dimensions (Steps to Reproduce, Observed Behavior, Expected Behavior). Treating the two forms as co-equal conditions, we convert both the human annotations and the LLM output into a common 1–5 ordinal reproducibility score with a single deterministic scoring function, and measure agreement with Cohen's kappa against a pre-registered threshold of $\kappa \geq 0.70$.

On the Raw reports the model reaches $\kappa = 0.578$ (95% bootstrap CI $[0.471, 0.684]$; accuracy 71.9%), a moderate but sub-threshold level of agreement, close to the $\kappa_{\text{control}} = 0.674$ between the two human annotators on the same reports. On the Improved reports, agreement falls to $\kappa = 0.335$ (CI $[0.226, 0.441]$; accuracy 62.6%), only fair and well below the human ceiling on those reports ($\kappa_{\text{control}} = 0.556$). The drop is driven by a significant optimistic bias that appears only on the Improved form: a Wilcoxon signed-rank test is non-significant on Raw ($p = 0.452$) but strongly significant on Improved ($p < 10^{-6}$), where the model rates 111 of 139 reports as reproducible (score ≥ 4) against the humans' 94, scoring higher than the annotator in 47 cases and lower in only 5. Across both forms, agreement is highest on the structurally checkable Steps to Reproduce dimension (Raw $\kappa = 0.693$, Improved $\kappa = 0.654$) and collapses on Observed Behavior (Raw $\kappa = 0.289$, Improved $\kappa = 0.130$). A coarse binarized check (score ≥ 4 as reproducible) remains high on both forms (87.1% and 86.3% accuracy). GPT-4o mini does not meet the pre-registered bar on either form, and its reliability degrades precisely on the AI-polished reports whose fluent surface it appears to over-trust — evidence that an LLM evaluator should be validated on the kind of text it will actually score, and is best used as a human-supervised coarse pre-filter rather than an autonomous evaluator.

Index Terms

large language models, unit test generation, mutation score, empirical study

II. Introduction

Bug reports are critical input to the software development process. However, without the ability to reproduce a bug, developers cannot diagnose or fix it. Recent work shows that fewer than 8% of bug reports on platforms like Jira meet the criteria for first-time reproducibility, creating a significant bottleneck in software maintenance. For large projects managing tens of thousands of reports monthly (such as TensorFlow or Django), automating the reproducibility evaluation step offers clear practical value.

Large language models (LLMs) like GPT-4o have shown promise in various software engineering tasks, including bug report analysis and improvement. Yet a critical gap remains: no prior work has measured whether an LLM can evaluate bug report reproducibility with sufficient agreement to developer judgment. Existing studies either use LLMs to generate or improve bug reports, or measure agreement between human raters alone—not between LLM evaluators and developer assessment.

This work addresses that gap by investigating whether an LLM-based reproducibility evaluator can achieve substantial agreement (Cohen's $\kappa \geq 0.70$) with manual human review. We use the ImproBR replication package [1], which is uniquely suited to the question because it contains two forms of the same 139 Mojira Minecraft bug reports: the original *Raw* reports as submitted, and *Improved* versions in which an LLM pipeline has rewritten each report to be more complete. Both forms were independently annotated by two human reviewers using the same rubric.

We treat the two forms as *co-equal conditions* and run the identical evaluation on each. This lets us ask not only whether GPT-4o mini agrees with human reproducibility judgment on ordinary reports, but also whether that agreement holds up on reports that another LLM has already polished. The two conditions are complementary: the Raw condition tests the model on the messy text a triage system actually receives, while the Improved condition tests whether the model can still track human judgment when the surface text has been made fluent and complete-looking by AI. A large gap between the two would be a warning that an LLM evaluator can be misled by fluent presentation rather than genuine reproducibility.

The contribution is three-fold: we establish a reusable measurement protocol for LLM-as-evaluator agreement on bug report quality; we provide an empirical baseline for whether LLMs can reliably assist in bug triage on both original and AI-improved reports; and we show that the two answers differ, which has direct implications for anyone considering an LLM evaluator in a pipeline that already contains LLM-generated text.

III. Related Work

Our study sits at the intersection of three lines of research: (i) empirical work characterizing what makes bug reports useful, (ii) techniques that assess and improve the *steps to reproduce* (S2R) that determine whether a report is reproducible, and (iii) the emerging use of large language models (LLMs) for bug report analysis and as automated evaluators. We review each theme and then position our contribution.

A. Bug Report Quality and Information Completeness

A long line of empirical research establishes that bug reports vary widely in quality and that this variation directly affects triage and resolution effort. Survey evidence from developers of large open-source projects shows a systematic mismatch between the information developers most need—steps to reproduce, stack traces, and test cases—and the information reporters typically provide [2]. Complementary quantitative studies model report quality from surface and textual features and confirm that such features predict how quickly a report is acted upon, framing quality assessment as a largely automatable task [3]. Broader content analyses across multiple projects further quantify how often key fields are missing and trace the downstream consequences of incomplete reports for developers who must recover the missing context [4]. Collectively, this theme motivates *reproducibility* as the quality dimension of greatest practical value, yet it treats quality mostly through proxy features rather than by judging whether a report is actually reproducible.

B. Reproducibility and the Quality of Steps to Reproduce

A second theme targets the S2R directly, since incomplete or ambiguous steps are the dominant obstacle to reproducing a failure. Early tool support augmented the reporting process itself, guiding Android reporters to produce more complete, reproducible steps at submission time [5]. Later work shifted toward automatically *assessing* S2R quality: Euler identifies the reported steps and flags missing or low-quality ones to give reporters actionable feedback [6], and more recent approaches combine language understanding with app UI models so that natural-language steps are grounded in concrete interface interactions [7]. Closest to our setting, ImproBR uses an LLM pipeline to detect and repair missing or ambiguous S2R, observed behavior, and expected behavior on the Mojira issue tracker, substantially raising the share of executable steps and fully reproducible reports [1]. A common thread, however, is that these S2R techniques are built for GUI/mobile applications and rely on program or dynamic analysis of a runnable app to establish the language-to-program mapping—an assumption that does not transfer to a game issue tracker such as Minecraft/Mojira, and none of them frames reproducibility assessment as a measured agreement task against human judgment.

C. Large Language Models for Bug Report Analysis and Evaluation

The third theme concerns LLMs applied to software engineering and, increasingly, as evaluators. Systematic reviews document the rapid adoption of LLMs across SE tasks and note that data quality and rigorous evaluation—not merely model or dataset scale—determine reliability [8]. In parallel, the "LLM-as-judge" paradigm studies when a model's assessments can substitute for human evaluation, together with the meta-evaluation methods and limitations that govern their trustworthiness [9]. This literature makes clear that an LLM used as an assessor must itself be validated against human annotators using an explicit reliability metric, rather than assumed correct.

D. Positioning of Our Work

Prior work either (i) characterizes bug report quality through proxy features without judging actual reproducibility, (ii) assesses S2R quality for GUI/mobile apps using program or dynamic analysis that is infeasible for the Minecraft/Mojira domain, or (iii) applies LLMs to *generate*, *improve*, or *classify* reports—including ImproBR on the same Mojira data [1]—without validating the LLM as a reproducibility *assessor*. Our study addresses this gap: we use an LLM to automatically assess bug report reproducibility on Mojira and evaluate it directly against human expert annotation, treating the LLM-as-judge concern seriously [9] by reporting Cohen's Kappa against a pre-registered threshold of $\kappa \geq 0.70$. Uniquely, we run this evaluation on *both* the original reports and the LLM-improved versions that ImproBR produces from them [1], treating the two as co-equal conditions, which lets us measure whether an LLM evaluator's agreement with humans holds up on text that another LLM has rewritten. To our knowledge, this is the first study to benchmark automated reproducibility assessment as a human-agreement task on this domain, and the first to quantify how that agreement changes on AI-improved reports.

IV. Methodology

A. Dataset

We used the RQ1 subset of the *ImproBR Replication Package*, publicly available on Figshare [1], [10]. This subset contains 139 real-world bug reports collected from Mojira, the official issue tracker for Minecraft, in *two forms*:

- **Raw**: the 139 reports in their original, as-submitted form;

- **Improved**: the same 139 reports after the ImproBR LLM pipeline rewrote each one to be more complete (clarifying or adding steps, observed behavior, expected behavior, and environment).

We treat the two forms as *co-equal conditions* and run the identical pipeline on each, because our research question concerns the model's agreement with humans on both original and AI-improved reports. Each of the 278 report instances (139 issues $\times$ 2 forms) was independently evaluated by two human annotators (Author 1 and Author 2) along three structured dimensions defined in the replication package's rubric (`evaluation_metrics.yaml`):

- **Steps to Reproduce (S2R)**: *Executable* or *Non-Executable*, with an irreproducibility category when applicable;
- **Observed Behavior (OB)**: presence (*Present*/*Not Present*) and, when present, sufficiency (*Sufficient*/*Insufficient*);
- **Expected Behavior (EB)**: presence and, when present, accuracy (*Accurate*/*Inaccurate*).

B. Deterministic Scoring Function

To obtain a single ordinal ground-truth score per report, we applied a fixed, deterministic scoring function to the structured labels of each annotator:

$$\text{score}(s2r, irr, ob, obl, eb, ebl) = \begin{cases} 5 & \text{if } s2r = \text{Exec.}, ob = \text{Suff.}, eb = \text{Acc.} \\ 4 & \text{if } s2r = \text{Exec.}, \text{otherwise} \\ 1 & \text{if } irr = \text{“wrong information”} \\ 3 & \text{if } ob \text{ and } eb \text{ both present} \\ 2 & \text{if only one of } ob, eb \text{ present} \\ 1 & \text{otherwise} \end{cases} \tag{1}$$

The same function was applied, separately for each form (Raw and Improved), to (i) the adjudicated consensus labels in `FinalResults.csv`, which reconciles the two independent annotators (Author 1 and Author 2) and serves as the ground truth for RQ1/SQ1–SQ3, and (ii) the structured labels extracted from the LLM's JSON output, so that any difference between human and LLM scores reflects a difference in how the report was judged rather than a difference in scoring logic. For each form we also report the inter-annotator agreement between Author 1 and Author 2's composite scores (before adjudication) as a human baseline (κ_{control}, Section IV-D). Applying the scoring function to the human ground truth confirms that ImproBR substantially raises reproducibility: the share of reports scored as executable (≥ 4) rises from 28.8% (40/139) on the Raw form to 67.6% (94/139) on the Improved form.

C. LLM Configuration

We used `gpt-4o-mini-2024-07-18` [11] with `temperature`=0, a fixed `seed`, and JSON-mode structured output, so that the model returns a label, quality assessment, and short rationale for each of the S2R, OB, and EB dimensions. Following the pre-registered protocol, we used a single fixed zero-shot prompt template, anchored to the rubric in `evaluation_metrics.yaml`, for all 139 reports of each form (one fixed template per form). The prompt supplies the report's summary, description, and available metadata, and instructs the model to judge only the report content, without access to ground-truth labels, annotator identities, or prior predictions. For each form, a 26-report pilot phase (Section IV-A) confirmed the zero-shot template as sufficient (κ between the LLM and the pilot ground truth of 0.567 on Raw and 0.576 on Improved, both above the 0.50 gate), so the pre-defined few-shot fallback was not triggered. All reports were scored in a single pass with the fixed prompt version, held identical between the pilot and the full run; we did not revise the prompt after inspecting evaluation results.

D. Statistical Analysis

All of the following analyses are run *separately for each form* (Raw and Improved), and our main comparison is between the two resulting sets of numbers.

RQ1/SQ1 (overall agreement). For each form, we computed Cohen's kappa [12] between the LLM's composite 1–5 score and the human ground-truth score across all 139 reports, together with a percentile bootstrap 95% confidence interval (2,000 resamples with replacement over reports). As a human baseline, we computed κ_{control} between Author 1 and Author 2's composite scores on the same reports of that form.

SQ2 (per-dimension agreement). We re-scored each dimension separately on a 1–3 scale (S2R: executable vs. not; OB: present-and-sufficient, present-only, or absent; EB: present-and-accurate, present-only, or absent) and computed Cohen's kappa independently for S2R, OB, and EB, applying the same rescoring function to both human and LLM labels.

SQ3 (disagreement analysis). We identified reports where the LLM and human composite scores differed and inspected the model's per-dimension rationale text alongside the corresponding human labels to characterize recurring disagreement patterns, following the four categories anticipated in the pre-registered protocol (missing/ambiguous steps, implicit vs. explicit expectations, unclear environment/version information, unstructured description).

TABLE I
OVERALL AGREEMENT, RAW VS. IMPROVED (N=139 EACH).

Measure	Raw	Improved
Composite κ (LLM vs. human)	0.578	0.335
95% bootstrap CI	$[0.471, 0.684]$	$[0.226, 0.441]$
Accuracy	71.9%	62.6%
κ_{control} (human–human)	0.674	0.556
Wilcoxon p (LLM vs. human)	0.452	$< 10^{-6}$
Binarized (≥ 4) accuracy	87.1%	86.3%
Meets $\kappa \geq 0.70$?	No	No

Bias and sanity checks. We ran a two-sided Wilcoxon signed-rank test ($\alpha = 0.05$) on the paired (human, LLM) composite scores to test for a systematic scoring shift, and, as a coarse sanity check, binarized both scores at the pre-registered cut (score $\geq 4 \Rightarrow$ "Reproducible") and report accuracy, precision, and recall for that binary task.

All statistics were computed with `scikit-learn` and `scipy` on the released ground-truth and LLM-output files for each form (`full_ground_truth_raw.csv` / `full_llm_output_raw.csv` and `full_ground_truth_improved.csv` / `full_llm_output_improved.csv`). The source code, prompt templates, and raw experimental outputs are available at https://github.com/LamSongToan/SWT301_SE1925_G6.

V. RESULTS

We report the identical analysis for the two co-equal forms, Raw and Improved, and our central finding is the contrast between them. Table I summarizes the headline agreement numbers.

A. Dataset Composition

Under the composite 1–5 scoring function (Eq. 1), the two forms have very different score profiles. On the **Raw** form the human ground truth is skewed toward low-to-mid scores: score 1 (7 reports), score 2 (70), score 3 (22), score 4 (24), score 5 (16). The LLM's Raw distribution is similar in shape (6, 75, 14, 23, 21). On the **Improved** form the human ground truth shifts strongly upward—score 1 (21), score 2 (1), score 3 (23), score 4 (22), score 5 (72)—consistent with ImproBR raising the executable share from 28.8% to 67.6%. The LLM shifts even further up on the Improved form (score 1: 13, score 2: 0, score 3: 15, score 4: 3, score 5: 108), assigning score 5 to 108 of 139 reports. Figures 1 and 2 show the two distributions.

B. RQ1/SQ1: Overall Agreement

On the Raw form, Cohen's kappa between the LLM's composite score and the human ground truth is $\kappa = 0.578$ (95% bootstrap CI $[0.471, 0.684]$), with accuracy 71.9%, macro-F1 0.666, macro-precision 0.695, and macro-recall 0.655. On the Landis and Koch scale this is in the upper half of the *moderate* band, just below *substantial*, and does not reach the pre-registered $\kappa \geq 0.70$ threshold. On the Improved form, agreement drops to $\kappa = 0.335$ (CI $[0.226, 0.441]$), accuracy 62.6%—only *fair* agreement. The two confidence intervals do not overlap, so the difference between the forms is itself statistically meaningful.

The drop is not merely an artifact of a harder ground truth. As human reference points we computed κ_{control} directly between Author 1 and Author 2's composite scores on each form: $\kappa_{\text{control}} = 0.674$ on Raw and $\kappa_{\text{control}} = 0.556$ on Improved. The human ceiling does fall on the Improved form—annotating polished reports, most of which look complete, is a more subjective judgment—but the LLM falls much further. On Raw the model sits close below the human ceiling (0.578 vs. 0.674, a gap of 0.096); on Improved it drops far below it (0.335 vs. 0.556, a gap of 0.221). Whatever makes the Improved reports harder to judge affects the model roughly twice as much as it affects a second human annotator.

Figures 3 and 4 show the confusion matrices. On Raw, disagreement is concentrated near the diagonal and the model rarely confuses the extremes. On Improved, the mass shifts into the bottom-right: the model concentrates its predictions on score 5 even where the human annotator assigned 3 or 4.

C. SQ2: Agreement by Dimension

Table II and Figure 5 report Cohen's kappa computed separately for each reproducibility dimension (S2R on its binary Executable/Non-Executable judgment; OB and EB rescored on their own 1–3 scales), for both forms.

The ranking of dimensions is the same on both forms—S2R strongest, then EB, then OB—but the two forms differ sharply in how much the softer dimensions degrade. Steps to Reproduce is the most robust: it stays near the human ceiling on both forms ($\kappa = 0.693$ on Raw, 0.654 on Improved), because it is close to a binary, checkable judgment. Expected Behavior and Observed Behavior, which require judging whether a free-text description is present and sufficient, hold up moderately on Raw (EB 0.556, OB 0.289) but nearly collapse on Improved (EB 0.161, OB 0.130). In other words, the polishing that ImproBR applies to the behavior descriptions is exactly where the model stops tracking human judgment: the rewritten text reads as complete, and the model credits it, even when the human annotator does not.

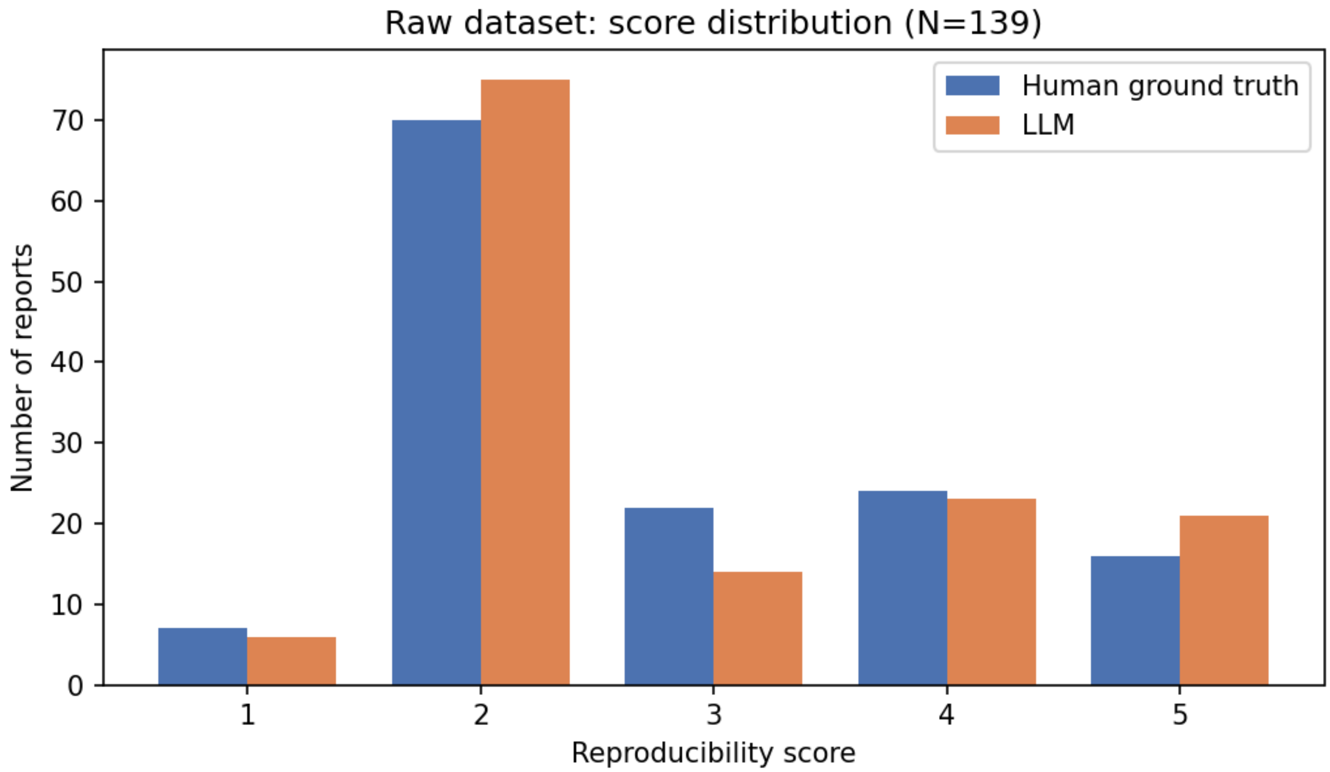

Fig. 1. Raw form: reproducibility score distribution, human vs. LLM (N=139).

TABLE II
COHEN'S KAPPA BY DIMENSION, RAW VS. IMPROVED (N=139 EACH).

Dimension	Raw κ	Improved κ
Steps to Reproduce (S2R)	0.693	0.654
Expected Behavior (EB)	0.556	0.161
Observed Behavior (OB)	0.289	0.130
Composite (RQ1)	0.578	0.335

D. Bias and Sanity Checks

The most informative contrast between the forms is in the bias test. On Raw, a two-sided Wilcoxon signed-rank test on the paired human/LLM composite scores is non-significant ($W = 337.5$, $p = 0.452$): the model shows no systematic shift, and its 39 disagreements (28.1%) split almost evenly into 21 over-scores and 18 under-scores. On Improved, the same test is strongly significant ($W = 158.5$, $p = 7.4 \times 10^{-7}$, rank-biserial correlation 0.89): of the 52 disagreements (37.4%), 47 are cases where the model scores *higher* than the human and only 5 are lower. The model develops a pronounced optimistic bias that exists only on the AI-improved reports—it rates 111 of 139 Improved reports as reproducible (score ≥ 4) against the humans' 94.

As a coarse sanity check, we binarized both scores at the pre-registered cut (score $\geq 4 \Rightarrow$ "Reproducible"). At this granularity the model performs well on *both* forms: Raw accuracy 87.1% (precision 75.0%, recall 82.5%) and Improved accuracy 86.3% (precision 83.8%, recall 98.9%). The Improved recall of 98.9% restates the optimistic bias from the other side: the model almost never misses a report the humans call reproducible, but only because it calls nearly everything reproducible. The gap between strong coarse performance and weak fine-grained ordinal agreement—mild on Raw, severe on Improved—recurs in Section VI.

E. SQ3: Qualitative Disagreement Analysis

On the **Raw** form we inspected the 17 reports where the human and LLM composite scores differ by two or more points, together with the model's per-dimension rationale text, and found two recurring, opposite-direction patterns.

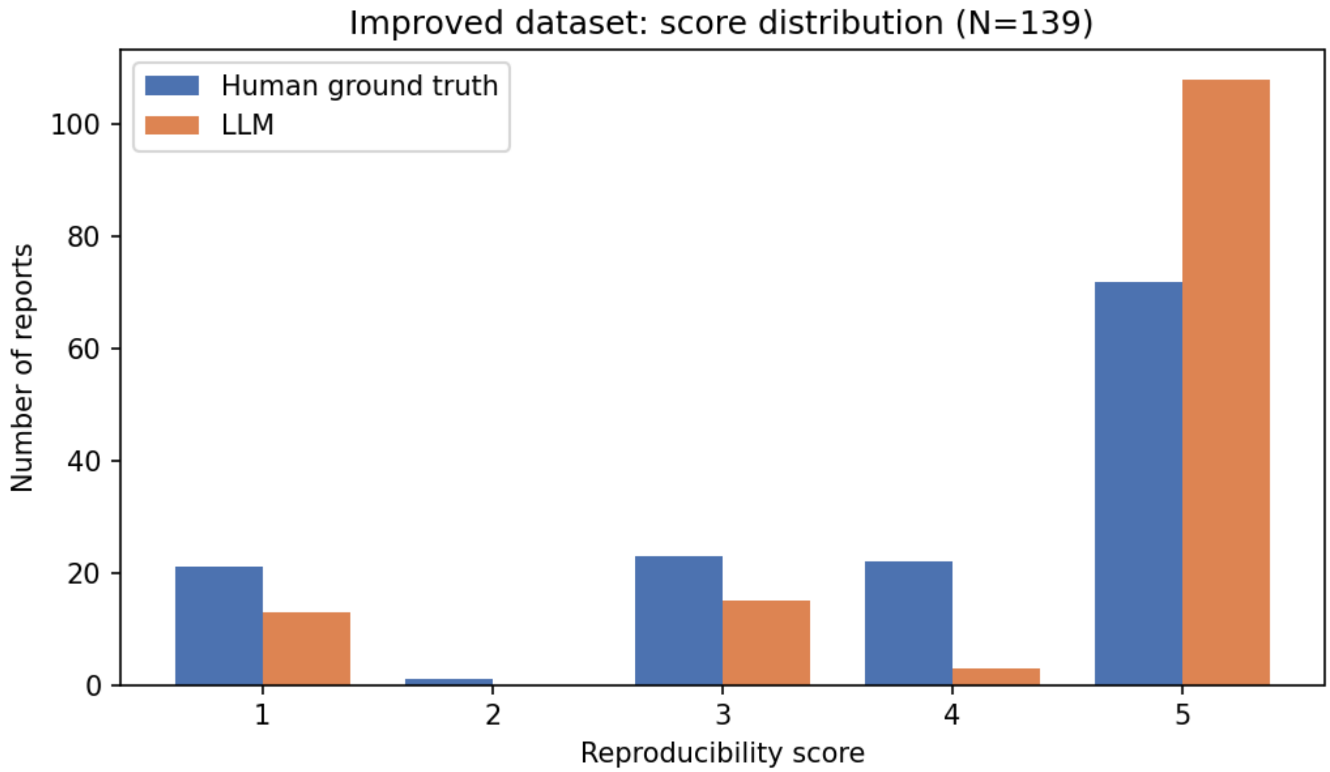

Fig. 2. Improved form: reproducibility score distribution, human vs. LLM (N=139).

LLM under-scores reports with implicit trigger conditions. In several cases (e.g., MC-300526, MC-300872, MC-301122) the human annotator judged S2R as *Executable* because the triggering condition could be inferred from the description, while the LLM required a more explicit, step-by-step trigger and labeled these reports *Non-Executable*, citing the absence of "a clear sequence of actions." This matches the *implicit vs. explicit* disagreement pattern anticipated in the study design.

LLM over-infers Expected Behavior from contextual cues. In the opposite direction (e.g., MC-300143), the human annotator marked Expected Behavior as *Not Present*, while the LLM inferred an expected behavior from contextual description and labeled it *Present/Accurate*. On the Raw form these two patterns roughly cancel, which is why the Wilcoxon test finds no net bias.

On the **Improved** form the second pattern dominates and the first almost disappears. Because the ImproBR rewrite fills in explicit steps and fluent behavior descriptions, the model rarely under-scores for missing structure any more; instead it takes the polished Observed and Expected Behavior text largely at face value and credits it as sufficient/accurate, where the human annotator still judges strictly and marks many as insufficient or inaccurate. This is why the over-scores outnumber under-scores 47 to 5 and why the OB and EB dimensions degrade most (Section V-C): the model's willingness to infer behavior from fluent text, which is a minor and self-cancelling error on Raw, becomes a systematic bias once every report is written to look complete.

VI. Discussion

A. The Threshold Is Missed on Both Forms, but for Different Reasons

Read strictly, the answer to the research question is negative on both forms: GPT-4o mini reaches $\kappa = 0.578$ on Raw and $\kappa = 0.335$ on Improved, neither meeting the pre-registered bar of $\kappa \geq 0.70$. But the two shortfalls have different characters. On Raw the gap is narrow and symmetric: the model sits just below *substantial* agreement and close to the $\kappa_{\text{control}} = 0.674$ measured between the two human annotators, and its errors do not lean in any direction. On Improved the gap is wide and one-sided: agreement is only *fair*, far below the human ceiling on that form (0.556), and driven by a significant optimistic bias. The Raw result is consistent with a capable-but-imperfect evaluator; the Improved result is consistent with an evaluator that is being misled.

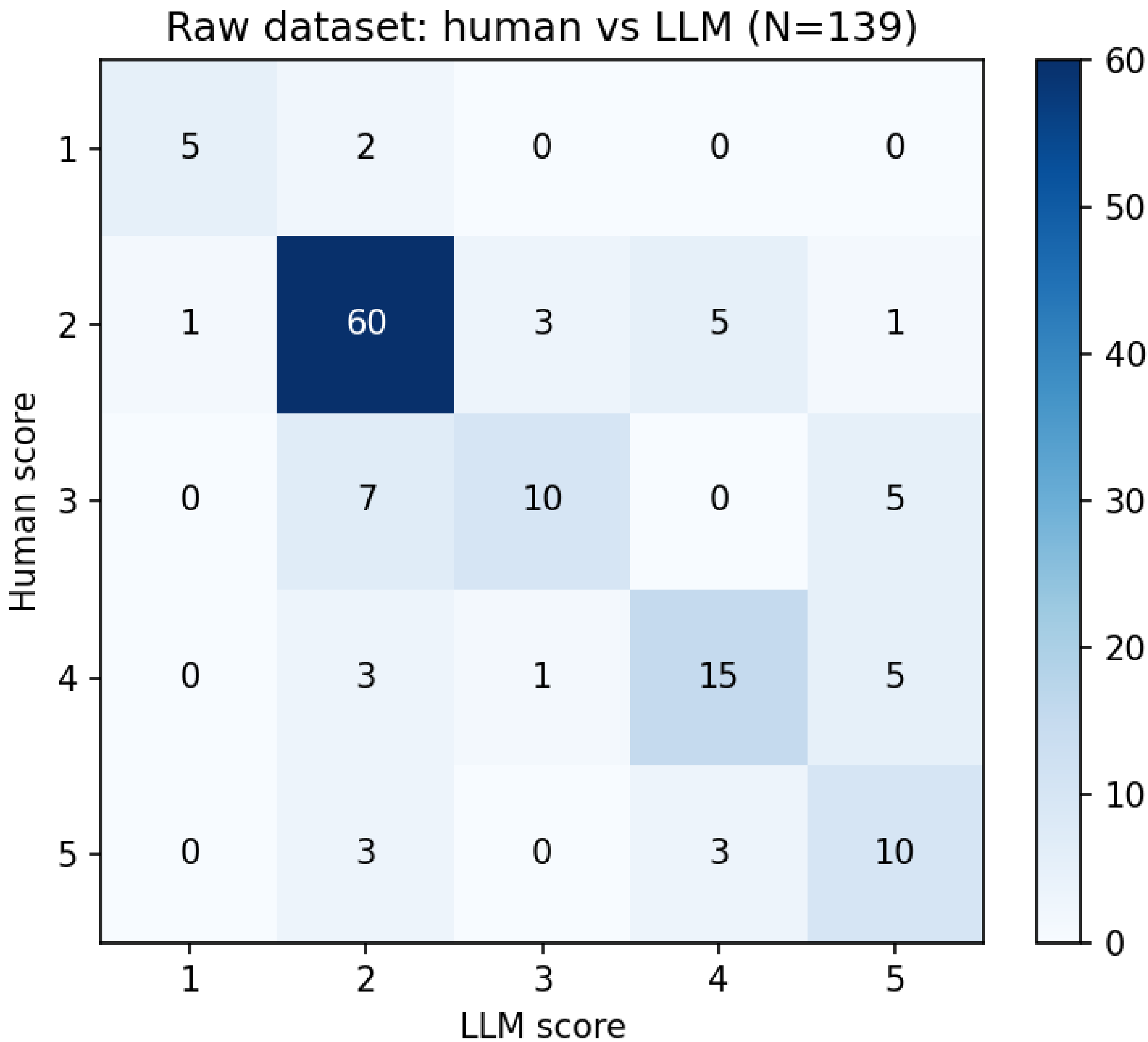

Fig. 3. Raw form: human vs. LLM composite scores (N=139).

B. The Central Finding: Agreement Degrades on AI-Improved Reports

The most important result of this study is the *difference* between the two forms. The same model, prompt, rubric, and scoring function, applied to the same 139 issues, agrees with humans far less well once those issues have been rewritten by another LLM. The degradation is concentrated exactly where one would expect if the model were responding to surface fluency rather than substance: it is small on Steps to Reproduce (a checkable, structural judgment that survives rewriting) and severe on Observed and Expected Behavior (free-text sufficiency judgments that the ImproBR rewrite makes read as complete). The bias test confirms the direction: on Improved the model scores higher than the human in 47 of 52 disagreements and calls 111 of 139 reports reproducible. In effect, the AI polish that makes a report look thorough also makes the evaluator credit it, whether or not the underlying reproducibility information is really there.

This has a concrete practical implication. Modern triage pipelines increasingly contain LLM-generated or LLM-rewritten text, and a natural design is to chain an LLM improver with an LLM evaluator. Our results warn that such a chain can be self-reinforcing: the evaluator systematically over-credits the improver's output. An LLM evaluator should therefore be validated on the specific kind of text it will score in deployment—validating it only on raw human reports would have overstated its reliability on the improved reports by nearly 0.25 kappa.

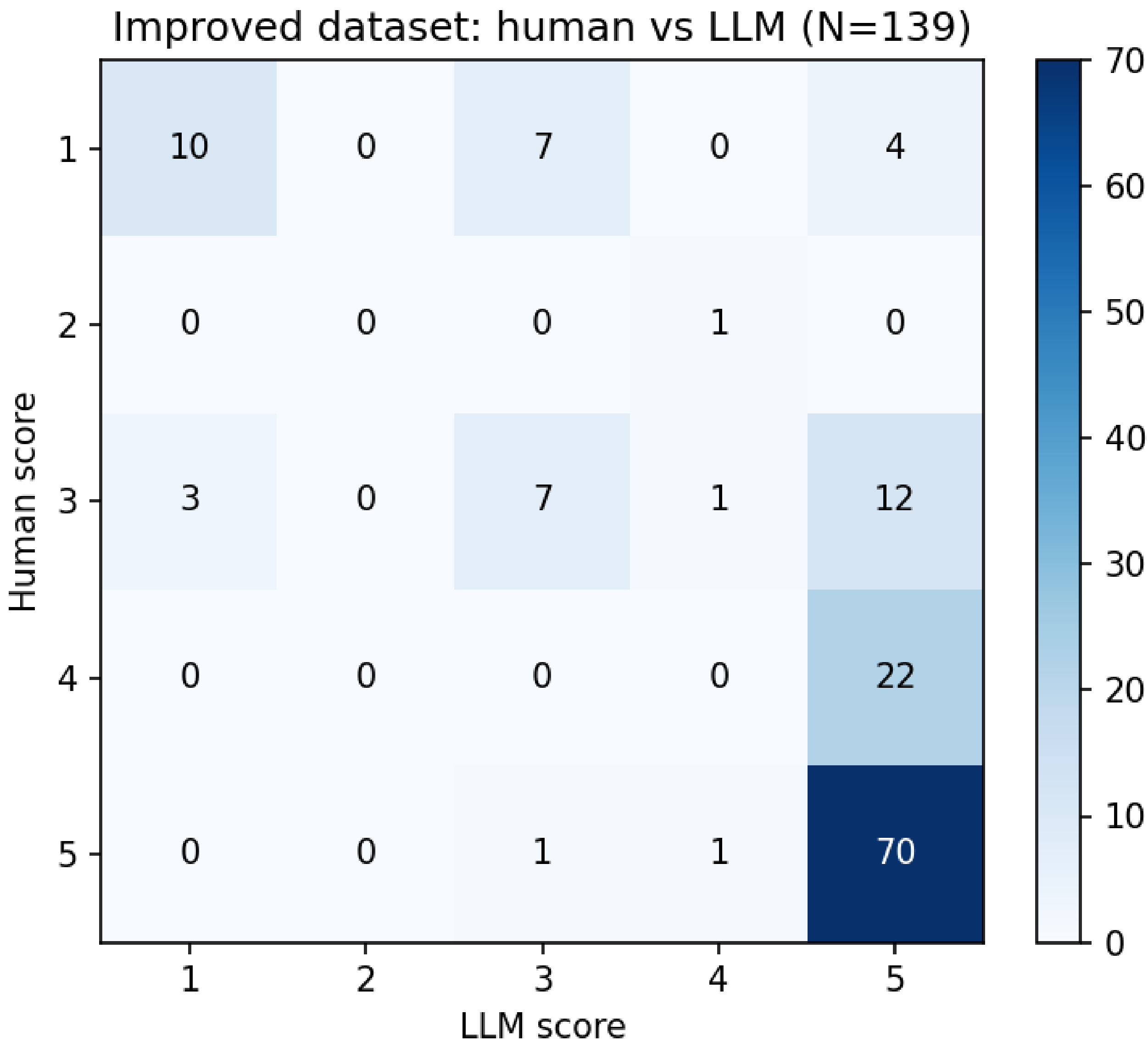

Fig. 4. Improved form: human vs. LLM composite scores (N=139).

C. Agreement Is Uneven Across Dimensions, Consistently

On both forms the dimension ranking is identical: Steps to Reproduce is strongest (Raw $\kappa = 0.693$, Improved 0.654), Expected Behavior is intermediate, and Observed Behavior is weakest (Raw 0.289, Improved 0.130). Steps to Reproduce is closest to a binary, checkable judgment—does the report state enough to attempt the action—while Observed Behavior sufficiency requires judging whether a free-text description is complete enough, a more subjective distinction. That the same ordering holds on both forms, while the softer dimensions degrade much more under rewriting, suggests an LLM-as-evaluator is more trustworthy for structurally checkable sub-judgments than for open-ended sufficiency judgments, and that this is a stable property of the model rather than an artifact of one dataset.

D. Coarse Screening Survives on Both Forms

The binarized check (score ≥ 4 as reproducible) reaches 87.1% accuracy on Raw and 86.3% on Improved, in both cases noticeably stronger than the fine-grained ordinal agreement. Most of the model's disagreement with humans happens between adjacent ordinal categories rather than between clearly reproducible and clearly not. Practically, GPT-4o mini is more dependable as a coarse first-pass filter than as a fine-grained 1–5 scorer—though on the Improved form the high binary recall (98.9%)

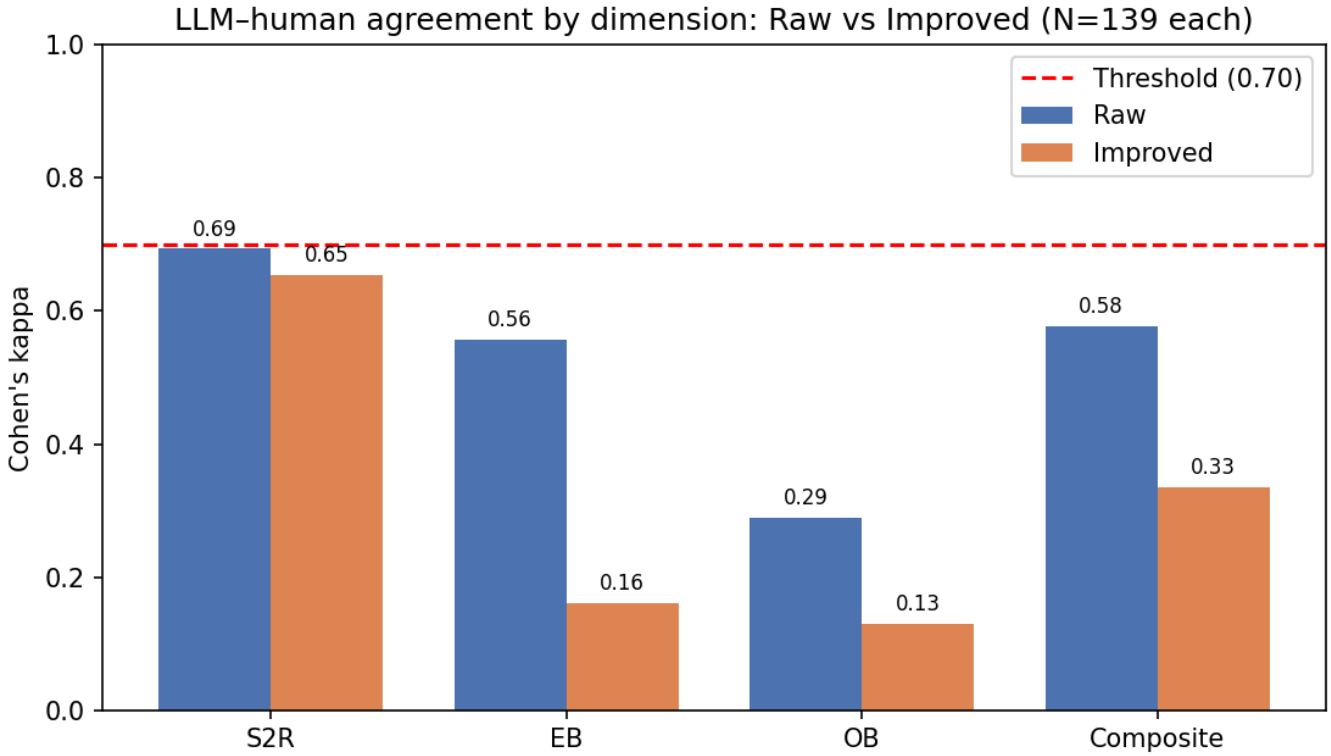

Fig. 5. Agreement by dimension, Raw vs. Improved, against the pre-registered threshold.

comes at the cost of precision, because the model calls almost everything reproducible, so even the coarse filter inherits the optimistic bias when the input has been AI-polished.

E. Relation to Prior Work

ImproBR reports that its pipeline raises the share of Executable reports from 28.8% to 67.6% on this dataset [1]; our human ground truth reproduces those exact figures, confirming that we are evaluating on the same material. Our contribution is complementary and cautionary: rather than improving reports, we ask whether an LLM can *judge* their reproducibility with human-level reliability, and we find that it approaches that bar on original reports but falls well short on the very reports ImproBR improves. Combined with the LLM-as-judge literature's caution that automated evaluators must be validated against human labels before their agreement figures can be trusted [9], our result sharpens that caution: validation must be done on the same distribution of text—including AI-generated text—that the evaluator will face in use, because agreement measured on human reports does not transfer to machine-rewritten ones.

VII. Threats to Validity

A. Internal Validity

The primary threat is *model non-determinism between runs or versions*. We mitigate this by using a fixed model snapshot (`gpt-4o-mini-2024-07-18` rather than a model alias), setting temperature to 0 for deterministic generation, and locking the prompt template to a single version throughout all 139 reports.

Another threat is *prompt design bias*, where the wording could inadvertently shift the LLM's scoring behavior. We reduce this through a clearly defined ordinal rubric (scale 1–5), and we use a controlled fallback strategy (zero-shot → few-shot only under predefined pilot conditions) rather than iterative prompt refinement.

B. Construct Validity

A key concern is whether we are truly measuring "reproducibility." Potential threats include misalignment between how humans and LLMs interpret reproducibility, and dependence on structured extraction (Steps to Reproduce, Expected/Observed Behavior) that may introduce abstraction bias.

We address this by anchoring the LLM rubric to the existing human annotation guidelines in the dataset, using the same ordinal scale for both human and LLM evaluation, and analyzing agreement separately for each reproducibility dimension (Steps, Observed Behavior, Expected Behavior) to reduce over-abstraction from a single composite score. Running the identical pipeline on both the Raw and Improved forms further guards against a construct threat specific to LLM evaluators: because the Improved form differs from Raw mainly in surface fluency rather than in the ground-truth construct, the sharp drop in agreement on Improved is evidence that the model partly responds to presentation rather than reproducibility itself—a threat we surface rather than hide.

C. External Validity

The study is limited to Mojira Minecraft bug reports, an open-source game development context. Results may not generalize to enterprise bug tracking systems or other domains with different reporting standards. The Improved form is also specific to one improvement pipeline (ImproBR); a different rewriting system might induce a different bias, though the mechanism we identify—an evaluator over-crediting fluent, complete-looking text—is not specific to ImproBR.

However, the dataset spans diverse technical bug types and is directly comparable to prior OSS studies, and evaluating both the original and the AI-improved form strengthens external validity relative to a single-form study: it shows the protocol applies to machine-rewritten reports and quantifies how much agreement changes between the two. The measurement protocol itself is domain-agnostic and can be replicated on other issue trackers and other improvement pipelines.

D. Conclusion Validity

Cohen's Kappa can be sensitive to class imbalance in ordinal scores, and bootstrap confidence intervals may fluctuate depending on resampling variance. The Wilcoxon signed-rank test assumes symmetric pair-wise differences.

We mitigate these by computing bootstrap confidence intervals for κ rather than relying on asymptotic assumptions, using non-parametric Wilcoxon testing (appropriate for ordinal data), and reporting effect sizes (rank-biserial correlation) alongside p-values.

E. Reproducibility

To ensure our results are reproducible, we fix the datasets to the 139 Raw and 139 Improved Mojira reports, pin the model version explicitly, lock one fixed prompt template per form (held identical between pilot and full run), remove randomness from sampling, and use standard statistical libraries (scikit-learn, scipy).

VIII. Conclusion

This study measures how closely an LLM-based reproducibility evaluator (GPT-4o mini) agrees with human judgment on the ImproBR replication package, which provides two co-equal forms of the same 139 Mojira Minecraft bug reports: the original *Raw* reports and their LLM-rewritten *Improved* versions. Using a single deterministic scoring function applied to both human annotations and LLM output, and Cohen's kappa against a pre-registered threshold of $\kappa \geq 0.70$, we run the identical evaluation on each form and compare.

On the Raw form the model reaches $\kappa = 0.578$—below the threshold but close to the $\kappa_{\text{control}} = 0.674$ between the two human annotators, with no systematic bias. On the Improved form agreement falls to $\kappa = 0.335$, far below the human ceiling on those reports (0.556), driven by a significant optimistic bias: the model scores higher than the human in 47 of 52 disagreements and calls 111 of 139 reports reproducible. The degradation is concentrated in the free-text Observed and Expected Behavior dimensions and is small on the structural Steps to Reproduce dimension (Raw $\kappa = 0.693$, Improved 0.654); a coarse reproducible/not check stays high on both forms (87.1% and 86.3%). The headline of the study is the contrast: the same evaluator that nearly matches human agreement on original reports is systematically misled by reports another LLM has polished.

Taken together, GPT-4o mini is not yet reliable enough to replace human reproducibility review on either form, and it is markedly less reliable on AI-improved reports than on original ones. It could realistically serve as a human-supervised coarse pre-filter today, but any deployment that scores LLM-generated or LLM-rewritten text must expect an optimistic bias and validate the evaluator on that kind of text specifically.

Future work should (i) target prompt or rubric refinement at the Observed and Expected Behavior dimensions, which are both the weakest and the ones most affected by the model crediting fluent text; (ii) test whether stricter grounding instructions or few-shot examples reduce the optimistic bias on AI-improved reports without harming Raw agreement; and (iii) replicate the protocol on additional issue trackers and additional improvement pipelines, to test whether the Raw-vs-Improved agreement gap observed here generalizes beyond Mojira/Minecraft and beyond ImproBR.

References

[1] E. F. Akyol, M. Dedeler, and E. Tüzün, "ImproBR: Bug report improver using LLMs," *arXiv preprint arXiv:2604.26142*, 2026. [Online]. Available: https://doi.org/10.48550/arXiv.2604.26142

[2] T. Zimmermann, R. Premraj, N. Bettenburg, S. Just, A. Schröter, and C. Weiss, "What makes a good bug report?" *IEEE Transactions on Software Engineering*, vol. 36, no. 5, pp. 618–643, 2010.

[3] P. Hooimeijer and W. Weimer, "Modeling bug report quality," in *Proceedings of the 22nd IEEE/ACM International Conference on Automated Software Engineering (ASE)*, 2007, pp. 34–43.

[4] S. Davies and M. Roper, "What's in a bug report?" in *Proceedings of the 8th ACM/IEEE International Symposium on Empirical Software Engineering and Measurement (ESEM)*, 2014.

[5] K. Moran, M. Linares-Vásquez, C. Bernal-Cárdenas, and D. Poshyvanyk, "Auto-completing bug reports for android applications," in *Proceedings of the 10th Joint Meeting of the European Software Engineering Conference and the ACM SIGSOFT Symposium on the Foundations of Software Engineering (ESEC/FSE)*, 2015, pp. 673–686.

[6] O. Chaparro, C. Bernal-Cárdenas, J. Lu, K. Moran, A. Marcus, M. Di Penta, D. Poshyvanyk, and V. Ng, "Assessing the quality of the steps to reproduce in bug reports," in *Proceedings of the 27th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE)*, 2019.

[7] J. Mahmud, A. Saha, O. Chaparro, K. Moran, and A. Marcus, "Combining language and app ui analysis for the automated assessment of bug reproduction steps," Proc. 33rd IEEE/ACM International Conference on Program Comprehension (ICPC), 2025. [Online]. Available: https://arxiv.org/abs/2502.04251

[8] X. Hou, Y. Zhao, Y. Liu, Z. Yang, K. Wang, L. Li, X. Luo, D. Lo, J. Grundy, and H. Wang, "Large language models for software engineering: A systematic literature review," 2024. [Online]. Available: https://arxiv.org/abs/2308.10620

[9] H. Li, Q. Dong, J. Chen, H. Su, Y. Zhou, Q. Ai, Z. Ye, and Y. Liu, "Llms-as-judges: A comprehensive survey on llm-based evaluation methods," 2024. [Online]. Available: https://arxiv.org/abs/2412.05579

[10] Anonymus, "ImproBR Replication Package," Figshare, Jan. 2026, software, Version 7. [Online]. Available: https://doi.org/10.6084/m9.figshare.30086083

[11] OpenAI, "GPT-4o mini: Advancing cost-efficient intelligence," OpenAI, Jul. 2024, accessed: Jul. 17, 2026. [Online]. Available: https://openai.com/index/gpt-4o-mini-advancing-cost-efficient-intelligence/

[12] J. Cohen, "A coefficient of agreement for nominal scales," *Educational and Psychological Measurement*, vol. 20, no. 1, pp. 37–46, Apr. 1960. [Online]. Available: https://doi.org/10.1177/001316446002000104